



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SATHYA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

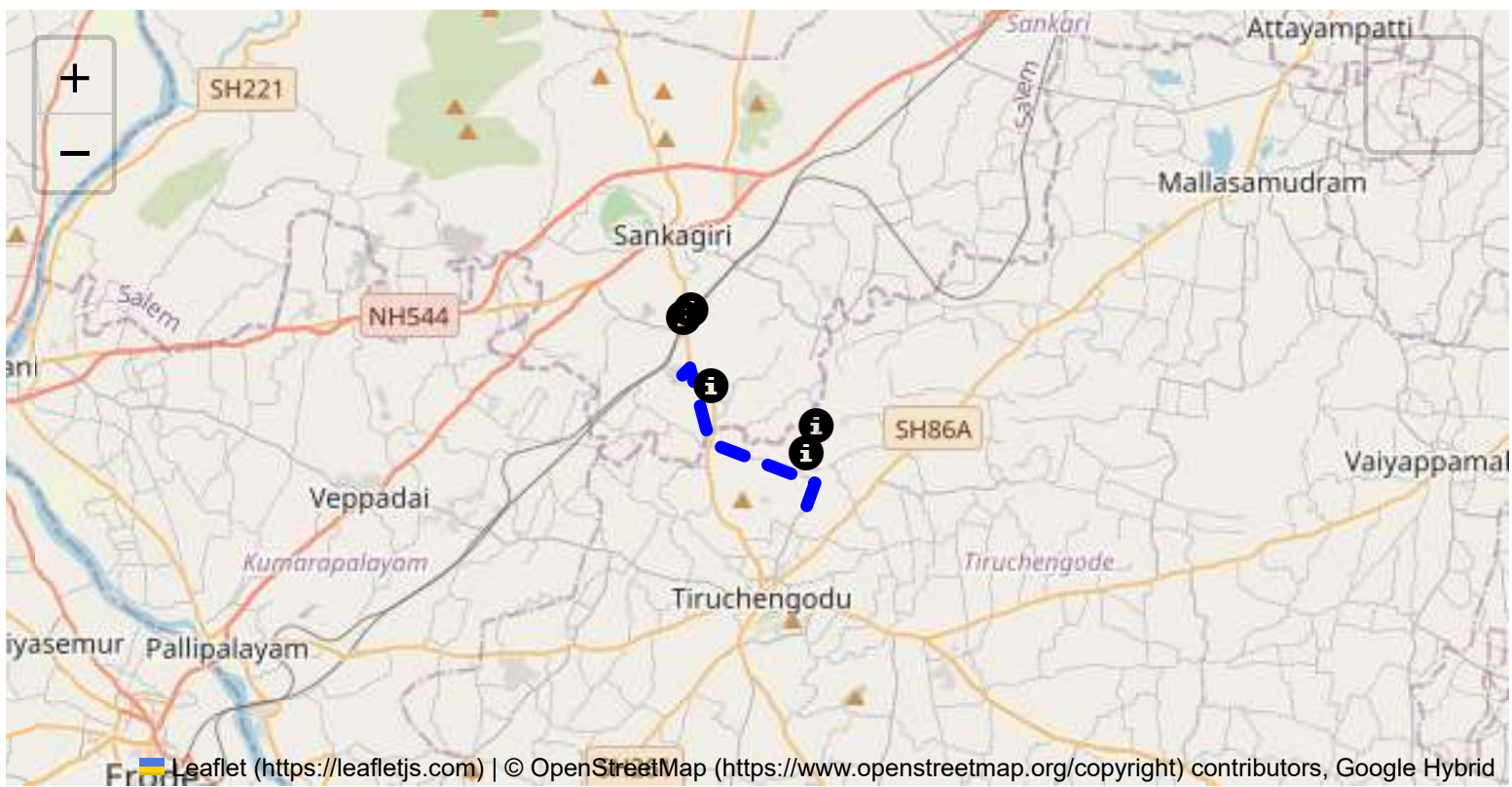
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

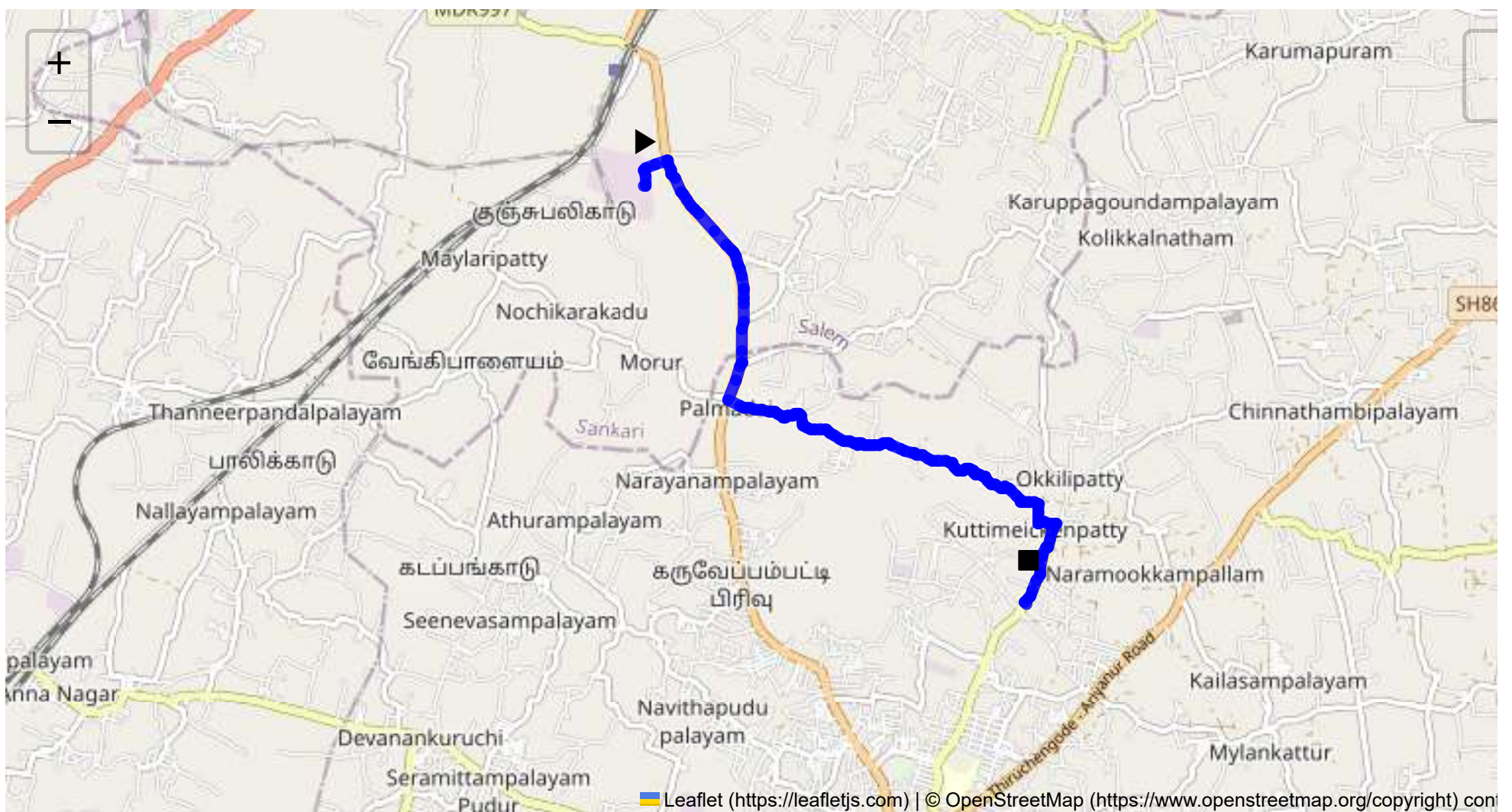
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 7.43 km
Estimated Duration: 0.2 hours
Adjusted Duration (Heavy Vehicle): 0.3 hours
Start: (11.4381, 77.8734)
End: (11.40221177, 77.90678111)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route spans approximately 7.43 kilometers. It starts from CVQF+23W, Sangagiri, traversing through CVRG+642, Kangeyampalayam, then moving onto CV9J+X8M, SH 86, Palmadai, and concluding at CW24+RPG, Kozhikkaalnatham Road. This route includes both rural roads and state highways.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region typically experiences a tropical climate with distinct wet and dry periods. Monsoon rains from June to September can lead to slippery roads and occasional flooding, while dry conditions can contribute to dust and reduced visibility. It's crucial to monitor weather forecasts during these periods for potential road hazards.

3. Analysis of Traffic Patterns

Traffic is generally moderate along this route. Peak hours occur in the mornings (7-9 AM) and evenings (5-7 PM) as locals commute to and from work. Congestion is typically more pronounced during these times, especially near marketplaces and junctions.

4. Assessment of Road Quality and Infrastructure

The overall road quality is moderate to good, with the state highway section (SH 86) being well-maintained. However, some rural sections might have potholes or rough patches. Ensure regular vehicle maintenance checks for optimal safety on these roads.

5. Suggestions for Alternative Routes for Emergencies

In case of emergencies, utilizing local roads connecting Sangagiri with nearby towns, like the roads through Bhavani and Tiruchengode, might serve as feasible alternatives, though they may increase travel time due to longer distances and potential slower traffic.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transportation of hazardous materials is subject to strict regulations. Ensure compliance with local laws, like clearly marked vehicles, appropriate load safety measures, and having necessary permits. Adherence to speed limits and restricted access zones is also critical.

7. Overview of Historical Incidents

While there's no widely reported history of major incidents specifically on this route, Tamil Nadu's history with road safety should prompt ongoing caution and preparedness, particularly in handling hazardous materials.

8. Environmental Considerations and Sensitive Areas

The route mostly passes through agricultural and semi-urban settings. Care should be taken near water bodies and farm areas to prevent contamination or accidents involving local wildlife.

9. Analysis of Communication Coverage

Cell coverage is generally reliable but may be spotty in more rural sections, particularly between Kangeyampalayam and Palmadai. It’s advisable to use a GPS system capable of offline routing.

10. Estimated Emergency Response Times

For urban segments, emergency response times are typically around 15 to 20 minutes. In rural sections, response times can extend to 30 or more minutes depending on the nature of the incident and traffic conditions.

11. Overall Summary of Risk Assessment

This route presents moderate risks due to potential weather effects during monsoon seasons and varying road quality. Careful attention to vehicle maintenance, compliance with local transport regulations, and monitoring traffic and weather reports are essential steps for ensuring a safe journey. Being proactive about emergency preparedness and having contingency plans for alternate routing will mitigate risks effectively.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Turn	High	11.41975, 77.88075	15 KM/Hr
5	Turn	Medium	11.41840, 77.88528	30 KM/Hr
6	Turn	Medium	11.41829, 77.88568	30 KM/Hr
7	Turn	Medium	11.41849, 77.88613	30 KM/Hr
8	Turn	Medium	11.41861, 77.88687	30 KM/Hr
9	Turn	Medium	11.41768, 77.88731	30 KM/Hr
10	Turn	Medium	11.41721, 77.88924	30 KM/Hr
11	Turn	Medium	11.41378, 77.90072	30 KM/Hr
12	Turn	Medium	11.41389, 77.90165	30 KM/Hr
13	Turn	Medium	11.41330, 77.90306	30 KM/Hr
14	Turn	Medium	11.41231, 77.90487	30 KM/Hr
15	Turn	Medium	11.41110, 77.90632	30 KM/Hr
16	Turn	High	11.41087, 77.90784	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
17	Turn	Medium	11.41083, 77.90786	30 KM/Hr
18	Blind Spot	Blind Spot	11.40927, 77.90787	10 KM/Hr
19	Blind Spot	Blind Spot	11.40911, 77.90939	10 KM/Hr
20	Blind Spot	Blind Spot	11.40243, 77.90667	10 KM/Hr

Risk Assessment - Elevation Changes

	Risk Type	Specific Risk	Risk Level	Coordinates	Speed Limit
0	Elevation Change	Ascent	Medium	11.41444754190697, 77.90010774942249	30 KMS/Hr

Route Photos of Risky Spots



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41975, 77.88075



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41840, 77.88528



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41829, 77.88568



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41849, 77.88613



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41861, 77.88687



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41768, 77.88731



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41721, 77.88924



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41378, 77.90072



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41389, 77.90165



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41330, 77.90306



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41231, 77.90487



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41110, 77.90632



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41087, 77.90784



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41083, 77.90786



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.40927, 77.90787



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.40911, 77.90939



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.40243, 77.90667
